

# SOURCES FOR RATIONAL TORSION SUBGROUPS ON GENUS-2 JACOBIANS OVER $\mathbb{Q}$ : DATED AND CLASSIFIED

For each torsion subgroup we list, in date order, the located sources that construct or exhibit it. *Inclusion rule:* the list aims to be exhaustive for explicit pre-2016 constructions and for proven geometrically simple realizations, and is representative (not exhaustive) for split constructions and for post-2016 database rows. A group has a section here only if it has some located literature beyond the databases; groups whose only witnesses are the production LMFDB, the extended Booker–Sutherland database, or the present paper (e.g. [63], [35], [45], [60], [2, 24], [3, 12] appearing inside the notes below) are not given their own section.

*Classification.* **geom. simple** = the source proves geometric (absolute) simplicity, or the displayed curve is independently certified so;  **$\mathbb{Q}$ -simple only**; **split** = explicitly split construction; **mixed** = a database or family row containing both geometrically simple and non-simple curves; **claimed** = simplicity asserted without proof; **not addressed**; **context** = a related result that is *not* itself a realization of  $\mathbb{Q}$ -rational torsion on a genus-2 Jacobian (a lead over a larger field, a torsion-free or Galois-stable-only example, an isogeny kernel, or a Prym rather than a Jacobian), retained for lineage. A \* marks a classification not verified against the original text (taken instead from a review, survey, or citing paper).

[5].

| Year  | Construction                                                                                              | Class                    | Reference / note                                            |
|-------|-----------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------|
| 1961* | $w^5=u(1-u)$ over $\mathbb{Q}(\mu_5)$ : rank 0 (a lead over $\mathbb{Q}(\mu_5)$ , not over $\mathbb{Q}$ ) | context                  | [16]                                                        |
| 1973* | $J_0(31)$ : cuspidal class of order 5 (exact by Mazur 1977)                                               | not addressed            | [41] (RM, geom. simple (LMFDB 961.a))                       |
| 1978* | Fermat-quotient cusp classes over $\mathbb{Q}(\zeta_5)$                                                   | context                  | [20]                                                        |
| 1986  | full torsion packet of $y^2-y=x^5$ , explicit                                                             | not addressed            | [12] (end of paper; the source cited by Boxall–Grant (0.2)) |
| 1991  | 2-param family, exact order 5 (Result 1, $g=2$ )                                                          | not addressed            | [18]                                                        |
| 2000  | torsion packet of $y^2-y=x^5$ ; $J$ CM by $\mathbb{Z}[\zeta_5]$                                           | geom. simple             | [6] (§5: absolutely simple (CM curve))                      |
| 2001  | universal 3-param family with marked 5-torsion                                                            | not addressed            | [3] (word simple absent from the paper)                     |
| 2004  | two 2-param 5-families (one contains $X_0(31)$ )                                                          | geom. simple             | [35] (generically absolutely irreducible)                   |
| 2016  | 725 exact-[5]: 650 RM/CM/split                                                                            | geom. simple, rest mixed | [5] (DB)                                                    |

[6].

| Year | Construction                                            | Class                    | Reference / note       |
|------|---------------------------------------------------------|--------------------------|------------------------|
| 1991 | 2-param family, exact order 6 (Result 1, $g=2$ )        | not addressed            | [18]                   |
| 2000 | split families with torsion containing $\mathbb{Z}/6$   | split                    | [22] (Table 1)         |
| 2016 | 595 exact-[6]: 378 RM/CM/split                          | geom. simple, rest mixed | [5] (DB)               |
| 2024 | exact $\mathbb{Z}/6$ on an O-PQM Jacobian (non-generic) | geom. simple             | [36] (Table 2, $D=6$ ) |

[7].

| Year  | Construction                                                                               | Class         | Reference / note                          |
|-------|--------------------------------------------------------------------------------------------|---------------|-------------------------------------------|
| 1973* | $J_0(29)$ : cuspidal class of order 7 (exact by Mazur 1977)                                | not addressed | [41] (RM, geom. simple (LMFDB 841.a))     |
| 2000  | $\mathbb{Z}/7 \times \mathbb{Z}/7$ , $\mathbb{Z}/63$ , $\mathbb{Z}/35$ split constructions | split         | [22] (Table 1)                            |
| 2004  | two 2-param 7-families (one contains $X_0(29)$ )                                           | geom. simple  | [35] (generically absolutely irreducible) |
| 2016  | 97 exact-[7]: 86 geom. simple, rest RM/split                                               | mixed         | [5] (DB)                                  |

[9].

| Year | Construction                                             | Class                     | Reference / note                             |
|------|----------------------------------------------------------|---------------------------|----------------------------------------------|
| 1991 | 1-param family, exact order 9 (Result 3, $g=2$ , $r=2$ ) | not addressed             | [18]                                         |
| 1995 | order-9 class inside the [3,9] family                    | $\mathbb{Q}$ -simple only | [32] (generic $\mathbb{Q}$ -simplicity only) |
| 2000 | split families containing $\mathbb{Z}/9$                 | split                     | [22] (Table 1)                               |
| 2016 | 30 exact-[9]: 29 geom. simple, 1 RM                      | mixed                     | [5] (DB)                                     |

[10].

| Year  | Construction                                                                                           | Class         | Reference / note               |
|-------|--------------------------------------------------------------------------------------------------------|---------------|--------------------------------|
| 1973* | $J_1(16)$ : torsion [2, 10] contains $\mathbb{Z}/10$ (geometrically split curve)                       | not addressed | [41] (cited in Leprévost 1995) |
| 1991  | 1-param family, exact order 10 (Result 3, $g=2$ , $r=1$ )                                              | not addressed | [18]                           |
| 2000  | split families containing $\mathbb{Z}/10$ ( $\mathbb{Z}/20$ , $\mathbb{Z}/30$ , $\mathbb{Z}/40$ , ...) | split         | [22] (Table 1)                 |
| 2016  | 131 exact-[10]: 126 geom. simple, rest split                                                           | mixed         | [5] (DB)                       |

[11].

| Year  | Construction                                                 | Class         | Reference / note                       |
|-------|--------------------------------------------------------------|---------------|----------------------------------------|
| 1973* | $J_0(23)$ : cuspidal class of order 11; exactness Mazur 1977 | not addressed | [41] (RM, geom. simple (LMFDB 529.a))  |
| 1977  | $J_0(23)(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/11$ exactly  | not addressed | [37] (Thm p.33 + Thm 1.2)              |
| 1990  | 1-param family with 11-torsion divisor (App. 3.1)            | not addressed | [17]                                   |
| 1991  | 1-param family, exact order 11 (Result 3, $r=0$ )            | not addressed | [18]                                   |
| 1995  | order-11 point inside AS $C_{22}$ (implicit)                 | geom. simple  | [32] (ambient Jacobian AS)             |
| 2009  | $X_0(23) + 17$ new curves; Flynn family generically AS       | geom. simple  | [4] (first explicit AS order-11 paper) |
| 2013* | uniform $f_{11}$ proven absolutely simple                    | geom. simple  | [47] (Platonov 2014 survey, Thm 7.4)   |
| 2016  | 8 exact-[11], all geometrically simple                       | geom. simple  | [5] (DB)                               |
| 2018  | CF family, exact order 11 (correction: arXiv v2 only)        | not addressed | [13] (correction in the arXiv v2 only) |

[13].

| Year  | Construction                                      | Class         | Reference / note                     |
|-------|---------------------------------------------------|---------------|--------------------------------------|
| 1990  | 1-param family with 13-torsion divisor (App. 3.2) | not addressed | [17]                                 |
| 1991* | 1-param family, order-13 class                    | not addressed | [27]                                 |
| 1991  | 1-param family, exact order 13 (Result 2, $r=0$ ) | not addressed | [18]                                 |
| 1995  | order-13 point inside AS $C_{26}$ (implicit)      | geom. simple  | [32]                                 |
| 2013* | uniform $f_{13}$ proven absolutely simple         | geom. simple  | [47] (Platonov 2014 survey, Thm 7.4) |
| 2016  | 7 exact-[13], all geometrically simple            | geom. simple  | [5] (DB)                             |

[14].

| Year  | Construction                                          | Class         | Reference / note                                     |
|-------|-------------------------------------------------------|---------------|------------------------------------------------------|
| 2012* | first order-14 curves $f_{14,1}, f_{14,2}$            | not addressed | [44]                                                 |
| 2013* | $J_{14,2}$ absolutely simple ( $J_{14,1}$ unresolved) | geom. simple  | [47] (Platonov 2014 survey, Thm 7.1)                 |
| 2015  | order-70 and order-28 curves contain 14 (implicit)    | split         | [23] (split by construction; no simplicity language) |
| 2016  | 12 exact-[14], all geometrically simple               | geom. simple  | [5] (DB)                                             |
| 2018  | point in Jsimple(2,14)                                | geom. simple  | [39] (Comp. 3.1.6)                                   |
| 2018* | infinite order-28 family contains 14 (implicit)       | not addressed | [42]                                                 |

[15].

| Year  | Construction                                                            | Class         | Reference / note |
|-------|-------------------------------------------------------------------------|---------------|------------------|
| 1991* | 1-param family, order 15                                                | not addressed | [28]             |
| 1992* | genus-g families incl. order 15                                         | not addressed | [29]             |
| 1996  | survey witness (Thm 1)                                                  | not addressed | [43]             |
| 2000  | $\mathbb{Z}/30, \mathbb{Z}/45, \mathbb{Z}/60$ split families contain 15 | split         | [22] (Table 1)   |
| 2016  | 17 exact-[15]: 10 geom. simple, rest split                              | mixed         | [5] (DB)         |
| 2018  | order-30 GS curve contains 15                                           | geom. simple  | [39] (Ex. 3.1.8) |

[16].

| Year  | Construction                                    | Class                     | Reference / note                           |
|-------|-------------------------------------------------|---------------------------|--------------------------------------------|
| 2001  | 32-family: $2P$ has order 16 (implicit)         | geom. simple              | [14] (earliest order-16 of any kind)       |
| 2013* | $\mathbb{Z}/48$ curves contain 16; proven split | split                     | [47] (Platonov 2014 survey, Thms 6.2, 7.2) |
| 2016  | 3 exact-[16], all geometrically simple          | geom. simple              | [5] (DB)                                   |
| 2018  | Jsimple(2,16)                                   | geom. simple              | [39] (Comps. 3.1.6, 3.1.12)                |
| 2024  | single order-16 curve                           | $\mathbb{Q}$ -simple only | [49] (Cor. 4.2 (v2))                       |

[17].

| Year  | Construction                              | Class         | Reference / note                     |
|-------|-------------------------------------------|---------------|--------------------------------------|
| 1991* | 1-param family, order 17                  | not addressed | [28]                                 |
| 1992  | 1-param family, order 17 (Thm 3.2.1)      | not addressed | [29] (full text verified)            |
| 1994* | family of order $2g^2+4g+1=17$            | not addressed | [31]                                 |
| 2001  | order-34 curves contain 17 (implicit)     | geom. simple  | [14]                                 |
| 2013* | uniform $f_{17}$ proven absolutely simple | geom. simple  | [47] (Platonov 2014 survey, Thm 7.4) |
| 2016  | 1 exact-[17] curve, geometrically simple  | geom. simple  | [5] (DB)                             |

[18].

| Year  | Construction                                          | Class                     | Reference / note                     |
|-------|-------------------------------------------------------|---------------------------|--------------------------------------|
| 2012* | first order-18 curves $f_{18,1}, f_{18,2}$            | not addressed             | [44]                                 |
| 2013* | $J_{18,2}$ absolutely simple ( $J_{18,1}$ unresolved) | geom. simple              | [47] (Platonov 2014 survey, Thm 7.1) |
| 2016  | 3 exact-[18], all geometrically simple                | geom. simple              | [5] (DB)                             |
| 2018  | Jsimple(2,18)                                         | geom. simple              | [39]                                 |
| 2024  | single order-18 curve                                 | $\mathbb{Q}$ -simple only | [49] (Cor. 4.2 (v2))                 |

[19].

| Year  | Construction                                                                           | Class         | Reference / note                                                                           |
|-------|----------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------|
| 1973* | order-19 point on $J_1(13)$ (discovery)                                                | not addressed | [41] (Mazur–Tate 1973)                                                                     |
| 1973  | $J_1(13)(\mathbb{Q}) = 19$ points exactly                                              | split         | [38] ( $\mathbb{Q}$ -simple AND geometrically split ( $M_2(\mathbb{Q})$ , Serre footnote)) |
| 1991* | 1-param family, order 19                                                               | not addressed | [28]                                                                                       |
| 1992  | 1-param family, order 19                                                               | not addressed | [29]                                                                                       |
| 2013* | uniform $f_{19}$ absolutely simple — FIRST proven-GS order 19; also a split $f_{19,1}$ | geom. simple  | [47] (Platonov 2014 survey, Thms 7.4–7.5)                                                  |
| 2018  | Jsimple(2,19)                                                                          | geom. simple  | [39]                                                                                       |

[20].

| Year  | Construction                                | Class         | Reference / note                          |
|-------|---------------------------------------------|---------------|-------------------------------------------|
| 1996* | split $\mathbb{Z}/20$ family (announcement) | split         | [21] (via Stoll’s review + MPIM preprint) |
| 1997  | 1-param family, order $2g(2g+1) = 20$       | not addressed | [33] (full text verified: no simplicity)  |
| 2000  | $\mathbb{Z}/20$ 2-param split family        | split         | [22] (Table 1)                            |
| 2001  | order-40 curve contains 20 (implicit)       | geom. simple  | [14]                                      |
| 2016  | 6 exact-[20]: 5 geom. simple, 1 split       | mixed         | [5] (DB)                                  |
| 2018  | Jsimple(2,20)                               | geom. simple  | [39]                                      |

[21].

| Year  | Construction                                                                    | Class         | Reference / note                        |
|-------|---------------------------------------------------------------------------------|---------------|-----------------------------------------|
| 1988* | $J_1(18)(\mathbb{Q}) = \mathbb{Z}/21$ (presumed; $X_1(18)$ geometrically split) | not addressed | [25]                                    |
| 1991* | 1-param family, order 21                                                        | not addressed | [28]                                    |
| 1992  | 1-param family, order 21 (Thm 3.2.5)                                            | not addressed | [29]                                    |
| 1993* | $C_{21}$ announced                                                              | not addressed | [30]                                    |
| 1995  | $C_{21}$ proven split (covers $E_1, E_2$ )                                      | split         | [32] (pp. 300-301)                      |
| 1997  | $J_0(26)$ cuspidal subgroup $\mathbb{Z}/21$ (split Jacobian)                    | split         | [11] (Thm 1(iii); full torsion in [10]) |
| 2000  | $\mathbb{Z}/21$ 2-param split family                                            | split         | [22] (§3.6)                             |
| 2016  | 388.a.776.1 geometrically simple (first certified GS [21])                      | geom. simple  | [5] (DB 2016/certification 2019)        |
| 2018  | Jsimple(2,21) — first paper-proof                                               | geom. simple  | [39]                                    |

[22].

| Year  | Construction                               | Class         | Reference / note               |
|-------|--------------------------------------------|---------------|--------------------------------|
| 1993* | $C_{22}$ announced                         | not addressed | [30] (via Ogawa’s citation)    |
| 1995  | $C_{22}$ absolutely simple, order-22 point | geom. simple  | [32] (p. 304, $D_4$ at $p=3$ ) |
| 2016  | 2 exact-[22], both geometrically simple    | geom. simple  | [5] (DB)                       |

[23].

| Year  | Construction                                                              | Class         | Reference / note                           |
|-------|---------------------------------------------------------------------------|---------------|--------------------------------------------|
| 1993* | $C_{23}$ announced                                                        | not addressed | [30]                                       |
| 1994  | family over $\mathbb{Q}(t)$ with order-23 divisor; infinitely many curves | claimed       | [40] (p.295: simplicity claimed, no proof) |
| 1995  | $C_{23}$ absolutely simple                                                | geom. simple  | [32] (p. 304)                              |
| 1997  | order-23 family generalizing Ogawa + new single curve                     | not addressed | [33] (Thm 3, Rem. 2)                       |
| 2013* | uniform $f_{23}$ absolutely simple                                        | geom. simple  | [47] (Platonov 2014 survey, Thm 7.4)       |
| 2018  | Jsimple(2,23)                                                             | geom. simple  | [39]                                       |

|      |                               |               |                                |
|------|-------------------------------|---------------|--------------------------------|
| 2024 | order-23 family (Thm 5.1/5.3) | not addressed | [49] (no simplicity statement) |
|------|-------------------------------|---------------|--------------------------------|

[24].

| Year  | Construction                                   | Class         | Reference / note             |
|-------|------------------------------------------------|---------------|------------------------------|
| 1993* | two non-isomorphic order-24 curves announced   | not addressed | [30] (Zbl 0783.14016)        |
| 1995  | $C_{24,1}$ absolutely simple; $C_{24,2}$ split | geom. simple  | [32] (pp. 304–305 / p. 301)  |
| 2000  | split families with $[2, 24]$ , $[2, 2, 24]$   | split         | [22] (Table 1)               |
| 2015  | order-24 families (glued)                      | split         | [23] (split by construction) |
| 2016  | 4 exact-[24]: 1 geom. simple, 3 split          | mixed         | [5] (DB)                     |
| 2018  | Jsimple(2,24) three ways                       | geom. simple  | [39]                         |

[25].

| Year  | Construction                                                          | Class                     | Reference / note                     |
|-------|-----------------------------------------------------------------------|---------------------------|--------------------------------------|
| 1993* | order-25 curve announced                                              | not addressed             | [30]                                 |
| 1995  | $C_{25}$ : $\mathbb{Q}$ -simple; author suspects geometric split-ness | $\mathbb{Q}$ -simple only | [32] (p. 303)                        |
| 2013* | $C_{25}$ proven geometrically split                                   | split                     | [47] (Platonov 2014 survey, Thm 7.3) |
| 2018  | Ex. 3.1.7: FIRST-EVER geometrically simple [25]                       | geom. simple              | [39] (Nicholls thesis, Ex. 3.1.7)    |

[26].

| Year  | Construction                                                    | Class         | Reference / note |
|-------|-----------------------------------------------------------------|---------------|------------------|
| 1993* | order-26 curve announced                                        | not addressed | [30]             |
| 1995  | $C_{26}$ absolutely simple                                      | geom. simple  | [32] (p. 305)    |
| 2015  | infinite 26-family from unpublished Leprevost preprint (report) | not addressed | [23]             |
| 2018  | Jsimple(2,26)                                                   | geom. simple  | [39]             |

[27].

| Year  | Construction                                  | Class         | Reference / note                    |
|-------|-----------------------------------------------|---------------|-------------------------------------|
| 1993* | order-27 curve announced                      | not addressed | [30]                                |
| 1995  | $C_{27}$ proven split (degree-3 cover)        | split         | [32] (pp. 301-302)                  |
| 2015  | three new 27-curves (search)                  | not addressed | [23] (simplicity absent from paper) |
| 2016  | 604.a.9664.2 geometrically simple             | geom. simple  | [5] (DB)                            |
| 2018  | own construction + classifies a Howe curve GS | geom. simple  | [39]                                |

[28].

| Year  | Construction                                      | Class         | Reference / note                          |
|-------|---------------------------------------------------|---------------|-------------------------------------------|
| 2012* | first order-28 curve (Thm 4)                      | not addressed | [44]                                      |
| 2013* | second 28-curve; $J_{28,1}$ GS / $J_{28,2}$ split | geom. simple  | [47] (Platonov 2014 survey, Thms 7.1–7.2) |
| 2015* | four new order-28 curves, absolutely simple       | geom. simple  | [46] (Platonov 2020 survey)               |
| 2015  | five new 28-curves via (3,3)-gluing               | split         | [23] (split by construction)              |
| 2016  | 249.a.6723.1 geometrically simple                 | geom. simple  | [5] (DB)                                  |
| 2018* | infinite family with order-28 points              | not addressed | [42]                                      |
| 2018  | Jsimple(2,28)                                     | geom. simple  | [39]                                      |

[29].

| Year  | Construction                                                      | Class         | Reference / note   |
|-------|-------------------------------------------------------------------|---------------|--------------------|
| 1993* | order-29 curve announced                                          | not addressed | [30]               |
| 1995  | $C_{29}$ absolutely simple                                        | geom. simple  | [32] (pp. 305-306) |
| 2014  | new $f_{29}$ via degree-29 unit; $\text{End} = \mathbb{Z}$ proven | geom. simple  | [48] (Thm 7.4)     |
| 2016  | 976.a.999424.1 geometrically simple                               | geom. simple  | [5] (DB)           |
| 2018  | Jsimple(2,29)                                                     | geom. simple  | [39]               |

[30].

| Year  | Construction                                     | Class         | Reference / note               |
|-------|--------------------------------------------------|---------------|--------------------------------|
| 1996* | 1-param order-30 family (preprint lost)          | not addressed | [34] (via HLP/Elkies/Nicholls) |
| 1996  | survey witness for order 30 over $\mathbb{Q}(t)$ | not addressed | [43] (Thm 1)                   |
| 2000  | $\mathbb{Z}/30$ 2-param split family             | split         | [22] (§3.5)                    |
| 2015* | order-30 curve, absolutely simple                | geom. simple  | [46] (Platonov 2020 survey)    |
| 2018  | Ex. 3.1.8 geometrically simple order-30 curve    | geom. simple  | [39]                           |

[31].

| Year | Construction                                                                                                                                                                                                                                                                                                                                                                        | Class         | Reference / note                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------|
| 2002 | rational order-31 <i>subgroup</i> without rational generator on $y^2 = 5x^6 - 4x^5 + 20x^4 - 2x^3 + 24x^2 + 20x + 5$ (Galois acts on the subgroup through $\{1, 5, 25\} \bmod 31$ )                                                                                                                                                                                                 | context       | [14] (Galois-stable only; no rational point of order 31)                                             |
| 2024 | curve over $\mathbb{Q}$ for the newform 1830.2.a.q (RM by $\mathbb{Q}(\sqrt{2})$ ), with the torsion order 31 of its isogeny class tabulated in the same repository                                                                                                                                                                                                                 | not addressed | [9] (old-data/label_to_curve_QQ.txt and modav-torsion.out; exact group and simplicity not addressed) |
| 2026 | GL2-type torsion algorithm: order 31 proven attained by some member of the 1830.2.a.q isogeny class (Thm 4.2/4.3); Conjectures 4.5/4.6 include 31 for $g = 2$ ; no curve equations given                                                                                                                                                                                            | context       | [8] (arXiv 2602.21047v3 + repo torsion_g2.txt)                                                       |
| 2026 | exact $J(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/31$ on the CEHJMPV curve, geometric simplicity ( $D_4$ certificate at $p = 11$ ), RM by $\mathbb{Z}[\sqrt{2}]$ — the first <i>explicit</i> order-31 realization, and the first on a genus-2 Jacobian (the existence of an order-31 point on some inexplicit member of this isogeny class already follows from the AC2026 row above) | geom. simple  | this paper (present paper; generator has irrational (conjugate-quadratic) Mumford support)           |

[32].

| Year | Construction                                | Class        | Reference / note                                    |
|------|---------------------------------------------|--------------|-----------------------------------------------------|
| 2001 | 1-param family + printed member             | geom. simple | [14] ( $D_4$ at $\ell=7$ ; page-wide AS convention) |
| 2018 | recovers Elkies + possibly-new second curve | geom. simple | [39] (Comp. 3.3.10)                                 |

[33].

| Year  | Construction                  | Class         | Reference / note                     |
|-------|-------------------------------|---------------|--------------------------------------|
| 2012* | first order-33 curve (Cor. 1) | not addressed | [45]                                 |
| 2013* | J33 absolutely simple         | geom. simple  | [47] (Platonov 2014 survey, Thm 7.1) |
| 2018  | own 33-curve, Jsimple(2,33)   | geom. simple  | [39]                                 |

[34].

| Year | Construction               | Class        | Reference / note               |
|------|----------------------------|--------------|--------------------------------|
| 2001 | two curves                 | geom. simple | [14] ( $D_4$ at $\ell=5, 17$ ) |
| 2018 | recoveries (3.1.11, 3.3.3) | geom. simple | [39]                           |

[36].

| Year  | Construction                                          | Class         | Reference / note                     |
|-------|-------------------------------------------------------|---------------|--------------------------------------|
| 2012* | first order-36 curve (Thm 2)                          | not addressed | [45]                                 |
| 2013* | second order-36 curve; both proven split              | split         | [47] (Platonov 2014 survey, Thm 7.2) |
| 2015* | order-36 curve, absolutely simple — FIRST simple [36] | geom. simple  | [46] (Platonov 2020 survey)          |

[39].

| Year  | Construction                                | Class         | Reference / note            |
|-------|---------------------------------------------|---------------|-----------------------------|
| 2002  | curve (2010 Lorenzini reduction correction) | geom. simple  | [14] ( $D_4$ at $\ell=5$ )  |
| 2015* | order-39 curve, absolutely simple           | geom. simple  | [46] (Platonov 2020 survey) |
| 2015  | one new 39-curve (search)                   | not addressed | [23]                        |
| 2016  | 1116.a.214272.1 (= Elkies' curve)           | geom. simple  | [5] (DB)                    |
| 2018  | own 39-curve                                | geom. simple  | [39]                        |

[40].

| Year | Construction                                                  | Class        | Reference / note            |
|------|---------------------------------------------------------------|--------------|-----------------------------|
| 2000 | split family with torsion containing $\mathbb{Z}/40$          | split        | [22] (§3.5)                 |
| 2001 | curve, 'highest known (6/2001) for a simple genus-2 Jacobian' | geom. simple | [14] ( $D_4$ at $\ell=17$ ) |
| 2018 | curve isomorphic to Elkies' + a second (simplicity open)      | geom. simple | [39] (Comp. 3.3.12)         |

[2, 10].

| Year  | Construction                                                        | Class         | Reference / note                          |
|-------|---------------------------------------------------------------------|---------------|-------------------------------------------|
| 1973* | $J_1(16)$ torsion = $[2, 10]$ (geometrically split curve)           | not addressed | [41] (p. 226, via Leprevost)              |
| 2012  | $J_1(16)(\mathbb{Q})_{\text{tors}} = [2, 10]$ , cuspidal            | not addressed | [24]                                      |
| 2016  | 7 exact- $[2, 10]$ : 5 geom. simple, 2 split ( $X_1(16)$ and 363.a) | mixed         | [5] (DB)                                  |
| 2024  | 2-level 5-torsion families contain $[2, 10]$                        | geom. simple  | [15] (generic $\text{End} = \mathbb{Z}$ ) |

[2, 12].

| Year | Construction                                 | Class | Reference / note |
|------|----------------------------------------------|-------|------------------|
| 2000 | split families containing $[2, 12]$          | split | [22] (Table 1)   |
| 2016 | 4 exact- $[2, 12]$ : 2 geom. simple, 2 split | mixed | [5] (DB)         |

[2, 2, 2, 2].

| Year  | Construction                                                                 | Class         | Reference / note                          |
|-------|------------------------------------------------------------------------------|---------------|-------------------------------------------|
| 1983* | fully split quintic (2-descent); $(\mathbb{Z}/2)^4$ automatic                | not addressed | [7] (lead only)                           |
| 1993* | two fully-split-quintic curves, $J(\mathbb{Q})$ computed                     | not addressed | [19]                                      |
| 1995  | rank $\geq 19$ absolutely simple Jacobians, but $J(\mathbb{Q})$ torsion-free | context       | [51] (not a torsion realization)          |
| 2000  | $[2, 2, 4, 8]$ -family contains $(\mathbb{Z}/2)^4$                           | split         | [22] (Table 1)                            |
| 2024  | $(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/5$ family (containing only, never exact) | geom. simple  | [15] (generic $\text{End} = \mathbb{Z}$ ) |

[2, 2, 2, 10].

| Year | Construction                                                       | Class        | Reference / note                                                                      |
|------|--------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------|
| 2024 | 2-dim family, torsion [2, 2, 2, 10] order 80; Clebsch–Klein moduli | geom. simple | [15] (generic End = $\mathbb{Z}$ ; ‘record for torsion on a simple genus-2 Jacobian’) |

[3, 3].

| Year  | Construction                                                                                                                                                                                                                 | Class        | Reference / note                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1988* | (3, 3)-split Jacobians; isogeny kernel, no rational torsion points                                                                                                                                                           | context      | [26]                                                                                                                          |
| 2000  | split families containing $(\mathbb{Z}/3)^2$ pointwise ([3, 9], [6, 6], [3, 12])                                                                                                                                             | split        | [22] (Table 1)                                                                                                                |
| 2002* | (3, 3)-split moduli; isogeny kernel only                                                                                                                                                                                     | context      | [50]                                                                                                                          |
| 2014  | Thm 6: rational parametrization of the POINTWISE $A(3, 3)$ chart; Ex. 19–21 are single curves with exact torsion $(\mathbb{Z}/3)^2$ , asserted absolutely simple (Ex. 19 misprint: $12x^3 - 105$ should read $13x^3 - 105$ ) | geom. simple | [2] (Acta Arith. 165; the generic (End = $\mathbb{Z}$ ) [3, 3] witness of the companion table is the database curve 5100.a.1) |
| 2016  | 8 exact-[3, 3] production curves, ALL split                                                                                                                                                                                  | split        | [5] (DB)                                                                                                                      |
| 2021  | $\sqrt{3}$ -level families with $\mathbb{Z}/3 \times \mu_3$ (Galois-stable, not pointwise rational)                                                                                                                          | context      | [1]                                                                                                                           |
| 2024  | exact $(\mathbb{Z}/3)^2$ on a PQM Jacobian                                                                                                                                                                                   | geom. simple | [36] (non-generic (QM))                                                                                                       |

[3, 6].

| Year | Construction                                                     | Class   | Reference / note   |
|------|------------------------------------------------------------------|---------|--------------------|
| 2000 | split families containing [3, 6]                                 | split   | [22] (Table 1)     |
| 2016 | 6 exact-[3, 6], all split                                        | split   | [5] (DB)           |
| 2024 | excluded for PQM Jacobians; realized on PQM Pryms, not Jacobians | context | [36] (Prop. 8.0.3) |

[3, 9].

| Year | Construction                                                | Class                     | Reference / note                     |
|------|-------------------------------------------------------------|---------------------------|--------------------------------------|
| 1995 | pointwise [3, 9] family (Thm 1.2.2), infinitely many curves | $\mathbb{Q}$ -simple only | [32] (absolute simplicity left open) |
| 1996 | survey witness                                              | not addressed             | [43]                                 |
| 2000 | split [3, 9] 2-param family                                 | split                     | [22]                                 |
| 2016 | 1 exact-[3, 9] curve, split                                 | split                     | [5] (DB)                             |

[6, 6].

| Year | Construction                                                               | Class | Reference / note |
|------|----------------------------------------------------------------------------|-------|------------------|
| 2000 | $\mathbb{Z}/6 \times \mathbb{Z}/6$ 2-param split family ( $\mathbb{P}^2$ ) | split | [22] (§3.4)      |
| 2016 | 196.a.21952.1 exact-[6, 6], split                                          | split | [5] (DB)         |

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